

The empirical recurrence for OEIS A297799: recovering a conjecture that is not written down

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2 September 2026

Abstract

OEIS A297799 records “Empirical recurrence of order 95”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 142$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 95, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 95 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture one cannot read

OEIS A297799 is “Number of $n \times 5$ 0..1 arrays with every 1 horizontally, diagonally or antidiagonally adjacent to 2, 3 or 5 neighboring 1’s.”. It has offset 1 and begins

1, 38, 293, 2195, 23781, 231921, 2174696, 21427001, ...

The entry states:

Empirical recurrence of order 95 (see link above).

As of the “Last modified” line on the live entry (Mar 10 2022 20:42:10, revision 6) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 142$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 142$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 284$ exact terms of the model returns order 95 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{95} c_i a(n-i),$$

c_1	7	c_{25}	2089013442935	c_{49}	−86744232741101307	c_{73}	468613504186747958
c_2	7	c_{26}	−7034152743937	c_{50}	213877054494768982	c_{74}	274961521834577040
c_3	434	c_{27}	30312572394552	c_{51}	397813845993246771	c_{75}	93510957468446828
c_4	−1821	c_{28}	−1360728681271	c_{52}	290988891568901288	c_{76}	−31650973210669216
c_5	−1104	c_{29}	42788973309498	c_{53}	−349372685632924699	c_{77}	−62053114484178952
c_6	−64886	c_{30}	−137104296952958	c_{54}	−963040234565558425	c_{78}	−58610869028885152
c_7	186687	c_{31}	−37473911822666	c_{55}	−824149366680897438	c_{79}	−41606512326847136
c_8	−2962	c_{32}	−191912166198760	c_{56}	277863442378781890	c_{80}	−19702403533791840
c_9	4650314	c_{33}	471820791371282	c_{57}	1698608412833529369	c_{81}	−1269077032864928
c_{10}	−10046690	c_{34}	280563977783043	c_{58}	1841929744935751909	c_{82}	1830005398733632
c_{11}	6244917	c_{35}	541103166735113	c_{59}	315005989023833013	c_{83}	762961512762368
c_{12}	−183986168	c_{36}	−1213128856345238	c_{60}	−1999534733019976629	c_{84}	516382526451712
c_{13}	315411798	c_{37}	−1109548200167786	c_{61}	−2968647320960027422	c_{85}	215902331745920
c_{14}	−319868758	c_{38}	−719337246783296	c_{62}	−1487176670569254374	c_{86}	63206433633280
c_{15}	4400322637	c_{39}	2961398141172792	c_{63}	1282001556387671095	c_{87}	1003153355264
c_{16}	−6071367261	c_{40}	2878797799093290	c_{64}	3015993342920039602	c_{88}	−2153459426304
c_{17}	7501829645	c_{41}	−2194373757940190	c_{65}	2344697336638403675	c_{89}	−1057653999616
c_{18}	−67770946438	c_{42}	−9273646941513091	c_{66}	159928355075088704	c_{90}	97590263808
c_{19}	72626272656	c_{43}	−7773512840030407	c_{67}	−1466701283524052564	c_{91}	51048349696
c_{20}	−102858975805	c_{44}	21608530560905401	c_{68}	−1750826791240558488	c_{92}	−13934329856
c_{21}	707603735690	c_{45}	35033015367364314	c_{69}	−927585556048410276	c_{93}	−4074766336
c_{22}	−528062570032	c_{46}	25119930721780835	c_{70}	−160051753091526033	c_{94}	1140326400
c_{23}	958393619559	c_{47}	−84999346252467047	c_{71}	334408045574809494	c_{95}	−62914560
c_{24}	−5297450598186	c_{48}	−130096604171982828	c_{72}	450638145154538248		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 95, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $95 + 4$ terms removed returns order 95 as well, so $\deg q_{\min} = 95 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 18 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $95 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 95 that this sequence satisfies — and that family turns out to have exactly one member.

References

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